analytical chemistry

Article

pubs.acs.org/ac

Peptide Fragmentation and Surface Structural Analysis by Means of ToF-SIMS Using Large Cluster Ion Sources

Yuta Yokoyama,[†] Satoka Aoyagi,[*,†] Makiko Fujii,[‡] Jiro Matsuo,[‡] John S. Fletcher,[§] Nicholas P. Lockyer,[∥] John C. Vickerman,[∥] Melissa K. Passarelli,[⊥] Rasmus Havelund,[⊥] and Martin P. Seah[⊥]

[†]Department of Materials and Life Science, Seikei University, Tokyo 180-8633, Japan

[‡]Quantum Science and Engineering Center, Kyoto University, Kyoto 611-0011, Japan

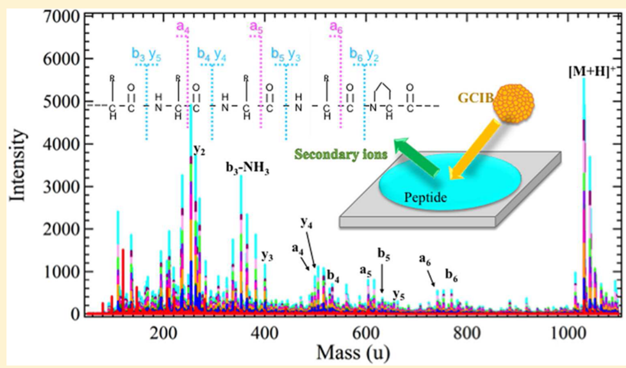
[§]Chemistry and Molecular Biology, University of Gothenburg, 40530 Göteborg, Sweden

[∥]Manchester Institute of Biotechnology and School of Chemistry, Manchester, M13 9PL, United Kingdom

[⊥]Surface and Nanoanalysis, National Physical Laboratory, Teddington, Middlesex, TW11 0LW, United Kingdom

Supporting Information

ABSTRACT: Peptide or protein structural analysis is crucial for the evaluation of biochips and biodevices, therefore an analytical technique with the ability to detect and identify protein and peptide species directly from surfaces with high lateral resolution is required. In this report, the efficacy of ToF-SIMS to analyze and identify proteins directly from surfaces is evaluated. Although the physics governing the SIMS bombardment process precludes the ability for researchers to detect intact protein or larger peptides of greater than a few thousand mass unit directly, it is possible to obtain information on the partial structures of peptides or proteins using low energy per atom argon cluster ion beams. Large cluster ion beams, such as Ar clusters and C_{60} ion beams, produce spectra similar to those generated by tandem MS. The SIMS bombardment process also produces peptide fragment ions not detected by conventional MS/MS techniques. In order to clarify appropriate measurement conditions for peptide structural analysis, peptide fragmentation dependency on the energy of a primary ion beam and ToF-SIMS specific fragment ions are evaluated. It was found that the energy range approximately $6 \leq E/n \leq 10$ eV/atom is most effective for peptide analysis based on peptide fragments and [M + H] ions. We also observed the cleaving of side chain moieties at extremely low-energy $E/n \leq 4$ eV/atom.



Since Ar cluster ions[1−3] were introduced into surface analysis such as time-of-flight secondary ion mass spectrometry (ToF-SIMS) they have been applied to many applications such as cleaning of complex samples[4] and organic depth profiling.[5] Moreover, it is very beneficial to analyze peptide samples using Ar clusters in terms of identification and structural evaluation.[6,7] Peptide structural analysis is essential for developing biodevices, medicines and diagnostic techniques. ToF-SIMS has advantages for peptide or protein structural analysis such as evaluating the orientation of immobilized peptides or proteins[8] because it has extremely high sensitivity for detecting the outermost surface (less than 2 nm). Conventional ToF-SIMS with a Bi cluster ion source generates mainly amino acid fragment ions from peptides or proteins but has also been reported to produce peptide molecular ions up to approximately 1600 u and to generate larger fragment ions above 300 u important for determining amino acid sequences.[9] In previous studies,[6,7,9] it has been reported that ToF-SIMS using Bi_3, C_{60} and large cluster ions such as Ar cluster ions provides peptide spectra similar to those generated by low-energy CID and allows for detection of large molecular ions of more than 1000 u and large fragment ions around 1000 u that are useful for amino acid sequence determination. Furthermore, peptide fragmentation dependence on the energy of a pulsed Ar cluster ion beam has also been reported[10] and it was suggested that peptide spectra obtained with a primary ion beam of $E/n \geq 10$ eV/atom (where E is the primary ion kinetic energy and n is the number of atoms in the cluster beam) are significantly different from those obtained with an Ar cluster ion beam of $E/n = 4$ eV/atom. Regarding the analysis of organic materials using ToF-SIMS, Seah et al.[11] reported that high mass peaks are dominant in the ToF-SIMS spectra obtained with Ar clusters below $E/n = 10$ eV/atom and Sheraz et al.[12] reported that fragmentation falls in the ToF-SIMS spectra obtained with water clusters below $E/n = 10$ eV/atom. According to Seah's 'universal equation' for Ar clusters,[11,13] it is also suggested that the relationship between sputtering yield and E/n changes at around $E/n = 10$ eV/atom in terms of organic material analysis.

Received: October 31, 2015
Accepted: February 26, 2016
Published: February 26, 2016

Moreover, Shen et al.[14] reported that the E/n value should be kept above but close to the threshold values (5 eV/atom in trehalose film analysis). Therefore, there may be an optimum E/n range for peptide structural analysis using fragment ions. To find the appropriate analysis conditions for peptide analysis, peptide spectra obtained using Ar clusters of $1.8 \leq E/n \leq 333$ eV/atom and C_{60} ion beams of 167−667 eV/atom were evaluated in this study. Peptides have repeated backbone structures, while they have a variety of side chains. Therefore, peptides can be good model samples to evaluate Ar cluster applications.

■ MATERIALS AND METHODS

Sample Preparation. Samples of peptides were prepared on two substrates; silicon wafer and indium tin oxide (ITO) coated glass. The silicon wafers were supplied by KYODO International, Inc. Kanagawa of Japan, and have a surface oxide layer of 10 nm, and the ITO coated glass substrates, with dimensions of 8 mm × 8 mm, were supplied by SIGMA-Aldrich Co., St Louis, MO, USA. The silicon wafer substrates were washed in isopropanol for 10 min using the sonic wave cleaner and then dried. The substrates were washed in isopropanol and ethanol using an ultrasonic bath and then washed in pure water using the ultrasonic bath.

Model peptides were YGGFL (Leu-enkephalin, MW 555, Scrum Inc.), [Val^5]-angiotensin I bovine (DRVYVHPFHL, M.W. 1281.7, Sigma-Aldrich), [Asn^1 Val^5]-angiotensin II (NRVYVHPF, MW 1030.5, Sigma-Aldrich), DRVYIHAF ([Ala^7]-angiotensin II, M.W. 1019, Scrum Inc., Tokyo), angiotensin II human (DRVYIHPF, MW 1045.5, Sigma-Aldrich), somatostatin (AGCKNFFWKTFTSC, MW 1638.7, Sigma-Aldrich), and somatostatin 28 (SANSNPAMAPRERKA-GCKNFFWKTFTSC, MW 3148.6). Moreover, ^{13}C and ^{15}N-labeled peptides, such as YGGF[$^{13}C_9$; ^{15}N]L (F residue-labeled Leu-enkephalin, MW 565, Scrum Inc.) and DRVYI[$^{13}C_6$; ^{15}N]HAF (I residue-labeled [Ala^7]-angiotensin II, MW 1027, Scrum Inc.), were measured. Each peptide was dissolved in pure water and then the solution was pipetted onto the cleaned substrates. The amount of each peptide on a substrate was several 100 ng/cm^2. The samples were dried in a vacuum desiccator and then the overviews of surfaces were checked by microscope before ToF-SIMS measurement. In addition, cleaned ITO substrates and silicon wafers without the peptides were also prepared as control samples.

ToF-SIMS Analysis. Positive ion spectra of each peptide sample were acquired with a ToF-SIMS J105 (Ionoptika Ltd., Hampshire), a TOF-SIMS.5 (Ion-Tof GmbH, Münster) and a ToF-SIMS Ar-GCIB oa-ToF MS developed by Kyoto University (not commercially available). The Ar-GCIB oa-ToF MS and J105 have a continuous Ar gas cluster ion beam as a primary ion, while the TOF-SIMS.5 has a pulsed Ar cluster, C_{60}^+ or C_{60}^{2+} ion beam as a primary ion. All measurements were performed at 500 μm × 500 μm as a raster size.

The measurements for angiotensin II, [Val^5]-angiotensin I, somatostatin and somatostatin 28 on Si substrates were performed with the J105. Regarding the Ar cluster ion source, Ar clusters ranging from 1 to >3000 atoms in size are available. In this measurements, 20 keV Ar_{60}^+ (333 eV/atom), 20 keV Ar_{500}^+ (40 eV/atom), 20 keV Ar_{1000}^+ (20 eV/atom), 20 keV Ar_{2000}^+ (10 eV/atom), and 40 keV C_{60}^+ (667 eV/atom) were used as the primary ions. The primary ion dose was 10^{12} ions/cm^2.

The measurements for [Val^5]-angiotensin I, angiotensin II, and somatostatin on ITO substrates were performed with TOF-SIMS.5. The primary beam current was 0.85 pA for C_{60}^+ (167 eV/atom) or 0.3 pA for C_{60}^{2+} (333 eV/atom). The primary ion dose was less than 10^{12} ions/cm^2 to ensure static conditions.

The measurements for Leu-enkephalin, (F residue-labeled)-Leu-enkephalin, [Ala^7]-angiotensin II, (I resilue-labeled)-[Ala^7]-angiotensin II, [Asn^1 Val^5]-angiotensin II and somatostatin on Si substrates were performed with Ar-GCIB oa-ToF MS. The average Ar cluster size was 1700 and the primary beam current was maintained at 40 pA. The acceleration energies are 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 17, and 20 kV (E/n = 1.8, 2.4, 2.9, 3.5, 4.1, 4.7, 5.3, 5.9, 6.5, 7.1, 8.2, 10.0, and 11.8 eV/atom, respectively).

■ RESULTS AND DISCUSSION

Peptide Analysis Using Ar Cluster Beams above E/n = 10 eV/atom and C_{60} Ion Beams. Figure 1 shows typical ToF-SIMS spectra of [Val^5]-angiotensin I obtained with 20 kV Ar_{2000}^+, 20 kV Ar_{500}^+, and 20 kV Ar_{60}^+ (E/n = 10, 40, and 333 eV/atom, respectively) and 40 kV C_{60}^+ (667 eV/atom). All of the spectra in Figure 1 were acquired under the same primary

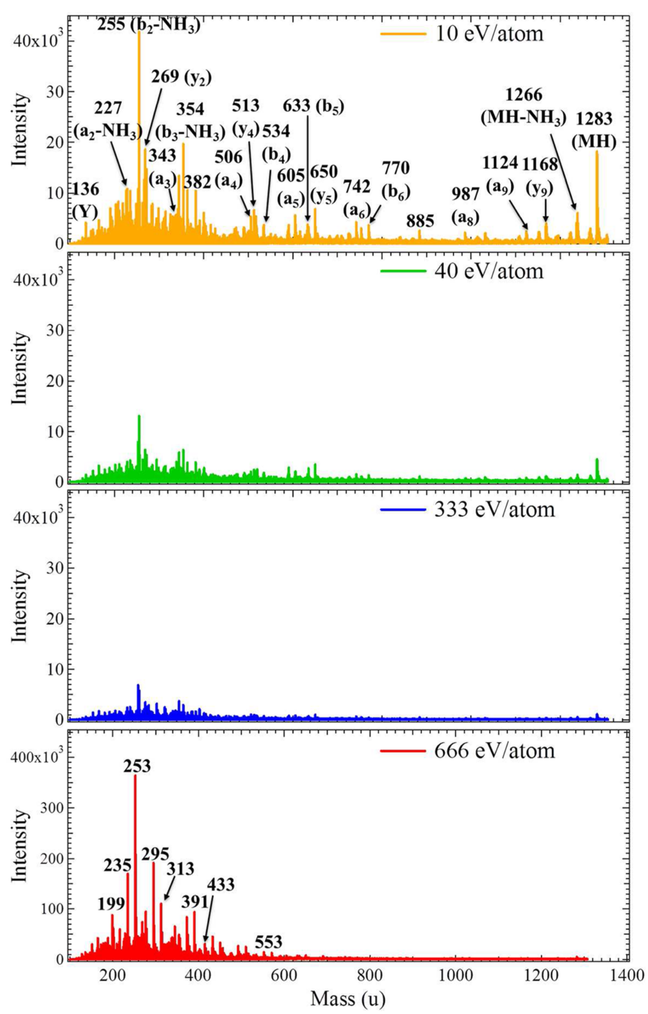


Figure 1. ToF-SIMS spectra of [Val^5]-angiotensin I bovine (DRVY-VHPFHL, MW 1281.7) obtained with 20 kV Ar_{2000}^+, 20 kV Ar_{500}^+, 20 kV Ar_{60}^+ and 40 kV C_{60}^+ primary ion beams.

DOI: 10.1021/acs.analchem.5b04133
Anal. Chem. 2016, 88, 3592−3597

ion dose density, 10^{12} ions/cm^2. The chemical fingerprint of [Val^5]-angiotensin I obtained with the C_{60} projectile was different than the chemical fingerprint obtained with the Ar cluster beam. The spectra obtained with the Ar cluster analysis beams show the same peaks regardless of the E/n values, however, the relative intensity of these peaks varied with the E/n-value. For peptide identification, the important secondary ions to detect are the molecular ions $[M + H]^+$ and fragment ions generally detected by low energy CID-MSMS, such as a-, b-, and y-type ions and amino acid fragment ions. In the previous study using a pulsed Ar cluster ion beam, the peptide spectra resulting from the primary ion energies from 10 to 40 eV/atom showed a similar tendency[9] though the intensities of peptide fragment ions obtained with E/n = 4 eV/atom are lower than those generated by above 10 eV/atom. In ToF-SIMS spectra of angiotensin II (DRVYIHPF, MW 1045.5) (shown in the Figure S1), obtained with different Ar cluster ion beams, the important peptide fragment ions and molecular ions are also strongly detected. In the spectra of E/n = 333 eV/atom, most of the important peaks are still available for identification or structural analysis though they showed the lowest intensity.

It has been reported that the fragmentation of a target organic molecule changes below E/n = 10 eV/atom.[10,11] For the analysis of organic materials using Ar clusters, it has been indicated that the slope of the relationship between sputtering yield and E/n changes approximately below 10 eV/atom.[10] Figure 2 shows an example of sputtering yield data (green circles) for Irganox 1010 and the fit using Seah's "universal equation"[10,12]

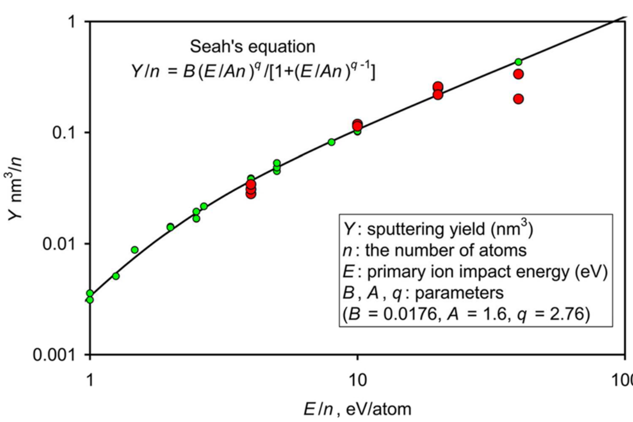


Figure 2. Example of sputtering yield data (green circles 2.5, 5, 10, 20 keV) and fit (solid black line) of eq 1 for Irganox 1010 from ref 13 together with scaled total ion yield volumes for [Val^5]-angiotensin I bovine (large red circles) from the present study.

$$Y/n = B(E/An)^q/\{1 + (E/An)^{q-1}\} \tag{1}$$

where Y is sputtering yield (nm^3), n is the number of atoms in the Ar cluster, E is primary ion impact energy (eV), and B, A, and q are parameters.[10] Since sputtering yield decreases with E/n, the spectrum pattern of peptides is expected to change at an extremely low E/n. This is generally observed for bulk organic materials as illustrated by Irganox 1010 with a similar molecular weight in Figure 2. Also shown in Figure 2 are data (large red circles) for the [Val^5]-angiotensin I bovine shown in Figure 1 deduced from the secondary ion spectra. We cannot measure the sputtering yields since data would have to be measured for thick layers of peptide which would have a different yield to these adsorbed layers.[15] Here we have summed the total measured ion yield but not just as a sum of the intensities but the product of intensity and mass to allow, approximately, for the different sizes of the ions.[11] This is then divided by the ion dose used and then scaled arbitrarily on the ordinate axis. It is clear that the behaviors are similar.

In the next section, peptide fragmentation at lower E/n is discussed. Moreover, not only the fragment ions generally detected with low energy CID but also ToF-SIMS specific peaks are detected by using Ar clusters. However, it is difficult to identify high mass unknown fragment ions. In terms of ToF-SIMS specific peaks, isotope-labeled peptides were used for supporting identification. This will be discussed in the last section.

In addition, ToF-SIMS spectra of somtatostatin 28 (Figure 3) suggest that a C_{60} primary ion is not always appropriate for obtaining large peptide secondary ions. In the case of [Val^5]-angiotensin I bovine, the spectrum by 40 keV C_{60}^+ is significantly different from those by Ar clusters as shown in Figure 1a. Smaller fragment ions have much stronger intensities than high mass peaks above approximately 600 u though the important large fragment ions that indicate peptide structures and [M + H] ions are detected strongly enough for identification. Moreover, Figure 4 shows ToF-SIMS spectra of [Val^5]-angiotensin I bovine obtained with C_{60}^+ and C_{60}^{2+}. In terms of secondary ion intensity, C_{60}^{2+} is slightly better than C_{60}^+ for detecting secondary ions larger than approximately 600 u.

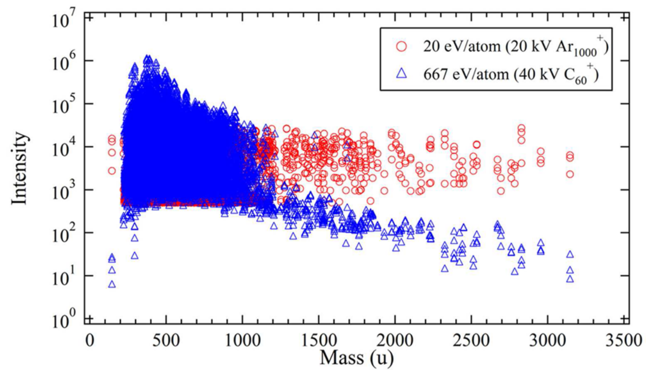


Figure 3. ToF-SIMS spectra of Somatostatin 28 (MW 3149) obtained with 20 kV Ar_{1000}^+ and 40 kV C_{60}^+ primary ion beams.

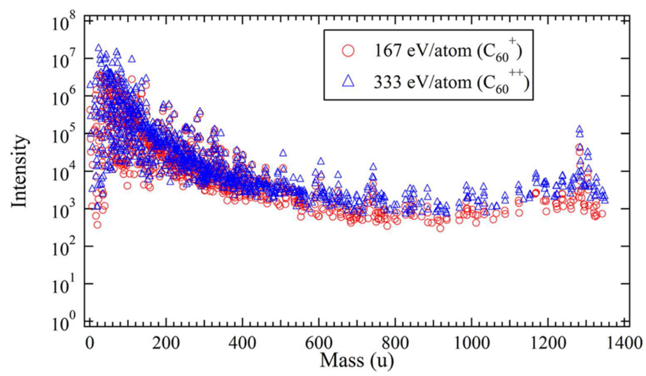


Figure 4. ToF-SIMS spectra of [Val^5]-angiotensin I (1281) obtained with 10 kV C_{60}^+ and C_{60}^{2+}.

DOI: 10.1021/acs.analchem.5b04133
Anal. Chem. 2016, 88, 3592−3597

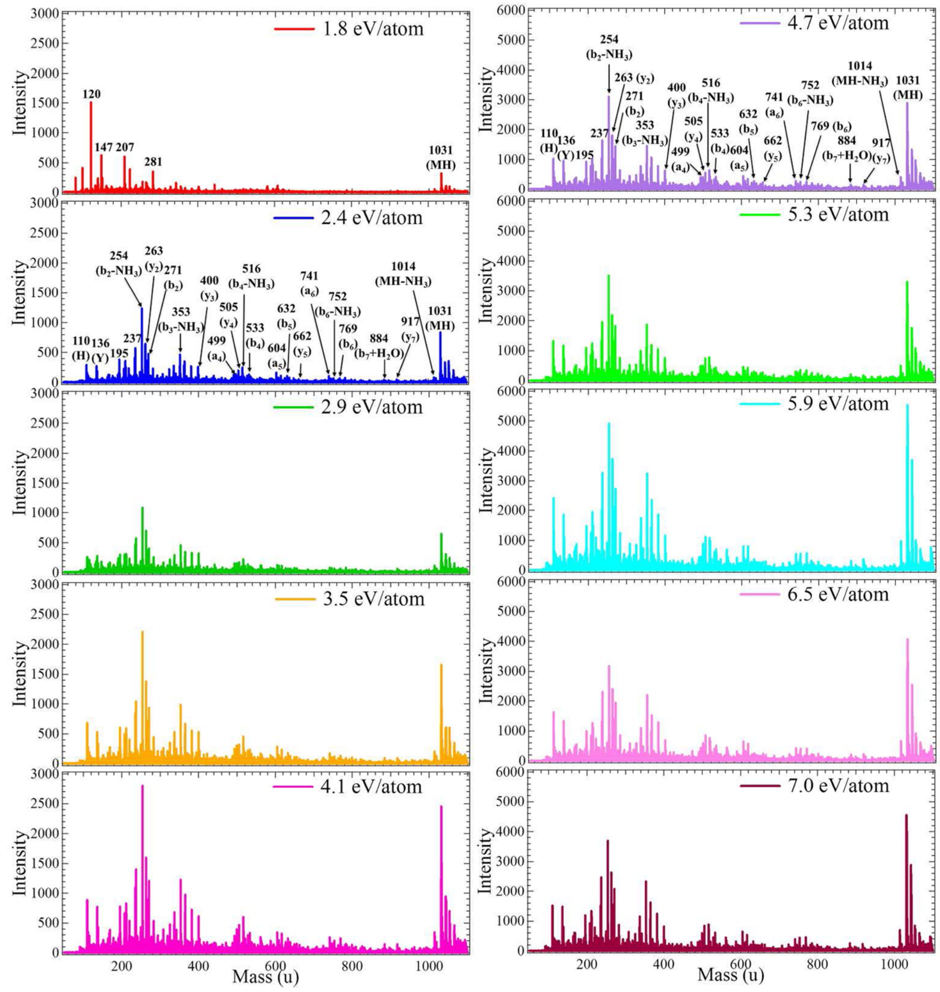


Figure 5. ToF-SIMS spectra of [Asn^1, Val^5]-angiotensin II (MW 1030) obtained with 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12 keV Ar_{1700}^+ (E/n = 1.8, 2.4, 2.9, 3.5, 4.1, 4.7, 5.3, 5.9, 6.5, and 7.1 eV/atom, respectively) primary ion beams.

Peptide Spectra Obtained with Ar Cluster Beams below *E/n* = 10 eV/atom. Figure 5 shows ToF-SIMS spectra of [Asn^1, Val^5]-angiotensin II (NRVYVHPF, M.W. 1030.5) obtained with Ar cluster ion beams of 1.8, 2.4, 2.9, 3.5, 4.1, 4.7, 5.3, 5.9, 6.5, and 7 eV/atom. Each energy measurement was conducted on the same sample area and the other sets of the spectra of two different areas are shown in Figure S4. The [Asn^1 Val^5]-angiotensin II spectrum of 1.8 eV/atom showed extremely low intensity for all secondary ions including [M + H]$^+$. In the spectrum by E/n = 1.8 eV/atom, polydimethylsiloxane (PDMS) peaks at m/z = 73 and 147 are highly detected, which indicates that ToF-SIMS using an Ar cluster ion beam of E/n below 2 eV/atom is so surface sensitive that it can detect only the outermost surface which is mainly covered by a contaminant such as PDMS. The spectra of the Si substrate were shown in the Figures S3 and S4.

Figure 6a shows the [M + H] ion intensity of YGGFL, labeled YGGFL, DRVYIHAF, and labeled DRVYIHAF obtained with 5, 7, 10, 14, 17, and 20 kV Ar_{1700}^+ (E/n = 3, 4, 6, 8, 10, and 12 eV/atom, respectively). Since it is suggested that an Ar cluster ion beam of E/n close to 1.8 eV/atom is too low to analyze peptides, Ar cluster ion beams of E/n = 3–12 eV/atom were employed. In the range $6 \leq E/n \leq 10$ eV/atom, the [M + H] ion intensity becomes highest. Figure 6b shows the intensity of an amino acid fragment ion (m/z 136, Y), a peptide fragment ion (m/z 506, a_4) and [M + H] ion (m/z 1028.5). The fragment ions also shows the highest intensity in the range $6 \leq E/n \leq 10$ eV/atom. Since the peptide related ions including [M + H]$^+$ and fragment ions are not stable below E/n = 5 eV/atom, it is indicated that a primary ion beam of $E/n \geq 5$ eV/atom is appropriate for peptide analysis including identification. The best E/n value for detecting the peptide fragment ions and [M + H]$^+$ would be between 6 to 10 eV/atom, though higher energy up to approximately several 100 eV/atom is effective.

Comparison of Peptide Spectra with ^{13}C and ^{15}N-Labeled Peptide Spectra. It has already been shown that peptide fragment ions indicating the amino acid sequence are significantly detected by large cluster ion beams, such as Ar clusters of E/n above 5 eV/atom. However, in ToF-SIMS spectra of peptides by Ar clusters some of the significant peaks are different from the peptide fragment ions generally detected by MSMS techniques. To clarify unknown secondary ions detected from peptide samples using Ar cluster ions, labeled peptide samples were analyzed. Sample peptides were Leu-enkephalin (YGGFL, MW 555.7), labeled Leu-enkephalin (YGG″F″L, all C and N atoms in F residue are ^{13}C and ^{15}N, respectively. MW 565.7), DRVYIHAF (MW 1020), and labeled DRVYIHAF (DRVY″I″HAF, all C and N atoms in I residue are ^{13}C and ^{15}N, respectively. MW 1027) and their ToF-SIMS

DOI: 10.1021/acs.analchem.5b04133
Anal. Chem. 2016, 88, 3592–3597

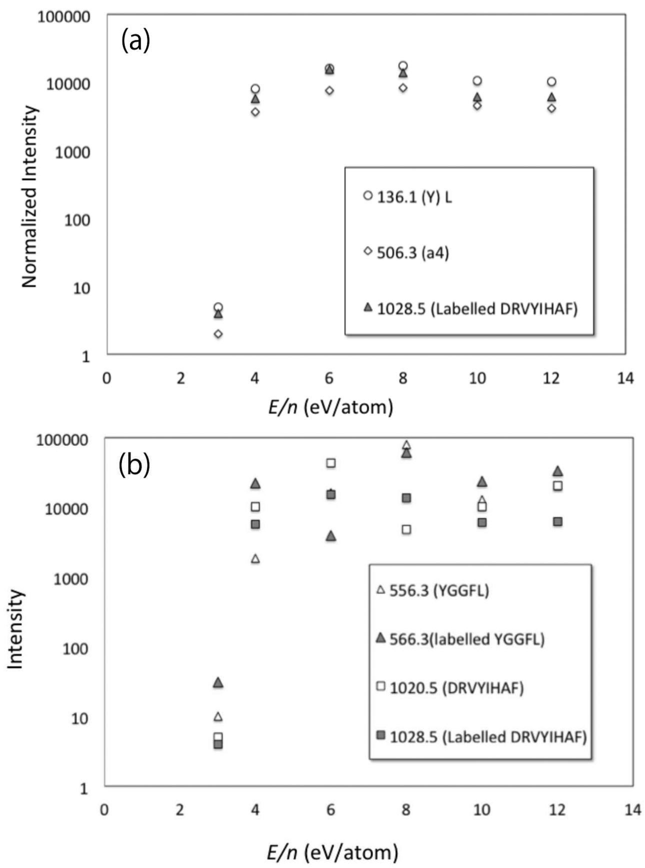


Figure 6. [M + H] ion intensity of YGGFL, labeled YGGFL, DRVYIHAF, and labeled DRVYIHAF obtained with 5, 7, 10, 14, 17, and 20 kV Ar_{1700}^+ (E/n = 3, 4, 6, 8, 10, and 12 eV/atom, respectively) primary ion beams (a), the intensity of an amino acid fragment ion (m/z 136, Y), a peptide fragment ion (m/z 506, a_4), and [M + H] ion (m/z 1028.5) (b).

spectra are shown in the Figures S4 and S5. DRVYIHAF has similar amino acid sequence to angiotensin II human (DRVYIHPF). Since proline has a ring structure including a peptide backbone, proline was replaced with one of the simple residues, alanine (A). In the spectra, there are peptide related secondary ions that are not generally detected by MSMS. Table 1 summarize significantly strong peaks in ToF-SIMS spectra of Leu-enkephalin and DRVYIHAF, respectively. Since there are so many peptide fragment ions Table 1 only contains some of the smallest and largest peaks and the rest of the peaks are shown in Tables S1 and S2. The secondary ions for the heavy isotope labeled peptides are 10 u (in the case of labeled YGG″F″L) or 7 u (in the case of labeled YGGFLDRVY″-I″HAF) larger than their unlabeled counterparts. Peptide fragments containing these heavy isotope moieties should also be shifted in the chemical fingerprint. According to the information regarding the inclusion of the labeled residues, unknown secondary ions were identified.

For instance, the large fragment ions 538.3, 510.2, 491.3, and 465.2 u from YGGFL and 548.3, 520.3, 491.2, and 475.2 u from labeled YGGFL show gradual fragmentation of YGGFL molecule as shown in Figure 7. Regarding DRVYIHAF peptide, it is indicated that a bond in the side chains could be cleaved by extremely low energy (Figure 7b). Thus, the origins of fragment ions suggested by the comparison to the labeled peptide were summarized in "suggested fragment" of Table 1.

As a result, it was found that bonds in amino acid side chains are often cleaved in ToF-SIMS measurement even using extremely low energy, approximately 4 eV/atom, primary ion beams. Moreover, it was shown that Ar cluster primary ions are useful for identifying peptides and evaluating peptide structures because they are able to produce important fragment ions and molecular ions, including large secondary ions higher than 3000 u, strong enough for identification when a primary ion beam of $E/n \geq 5$ eV/atom is used.

■ CONCLUSIONS

Ar cluster ion beams are useful for peptide analysis including identification and structural evaluation based on peptide fragment ions and $[M + H]^+$. The most appropriate energy range of Ar clusters for peptide analysis is $6 \leq E/n \leq 10$ eV/atom. The primary ion of $E/n < 2$ eV/atom is not appropriate for peptide analysis because even the $[M + H]^+$ ion is not clearly detected. By comparing the spectra of nonlabeled and isotope labeled peptides, cleavages by low energy Ar cluster ion beams are suggested. Even under E/n = 4 eV/atom, bonds in side chains could be cleaved.

Table 1. Main Secondary Ions Detected from YGGFL, Labeled YGGFL, DRVYIHA, and Labeled DRVYIHAF Samples[a]

mass (u)				mass (u)			
YGGFL	YGG″F″L	MSMS fragment	suggested fragment	DRVYIHAF	DRVY″I″HAF	MSMS fragment	suggested fragment
120.1	***129.2***	F		***905.5***	***912.5***	y7	
279.2	***289.3***	y2		***927.4***	***934.5***		MH-C_6H_4OH (from Y)
336.2	***346.2***	y3		***943.4***	***950.4***		MH-C_6H_5 (from F)
465.2	***475.2***		M-90 M-$(CH_2(CH_3)_2)$-COOH-2H	***959.4***	***966.5***		M-CH_3COOH (from D)
278.1	278.1	b3		***965.3***	***972.4***		MH-CN_2H (from R)
279.2	***289.3***	y2		***974.5***	***981.6***		MH-COOH (from D or F)
491.3	***491.2***		M-64 M-CH_3-COOH-4H	***990.5***	***997.5***		M-CH_2-NH
510.2	***520.3***	MH	M-COOH	***1003.5***	***1010.5***	MH-NH3	
538.3	***548.3***		M-OH	***1007.4***	***1014.4***		M-C
556.3	***566.3***			***1020.5***	***1037.6***	MH	

[a][M-90] means atoms of 90 u are removed from the molecule. Ions indicated in bold-italic contain the labeled residue.

DOI: 10.1021/acs.analchem.5b04133
Anal. Chem. 2016, 88, 3592−3597

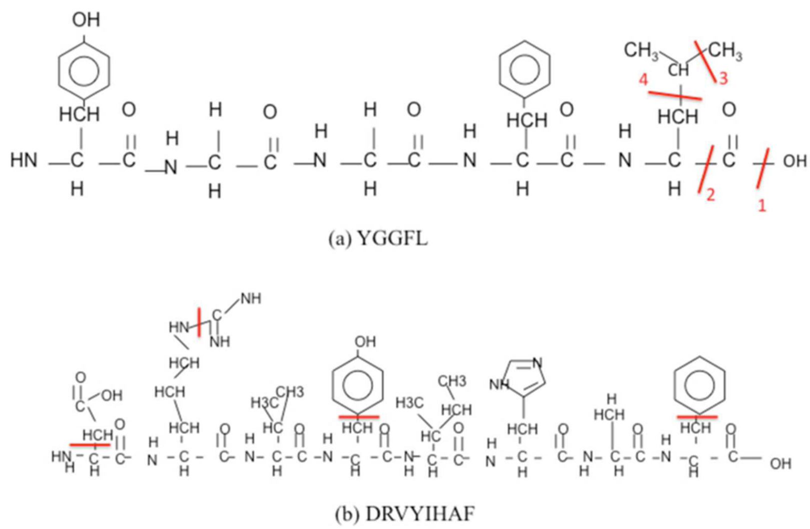


Figure 7. Suggested cleavages by Ar cluster ion beams: (a)YGGFL and (b)DRVYIHAF.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.analchem.5b04133.

ToF-SIMS spectra of angiotensin II (Figure S1), Si substrate (Figure S2), the magnified spectra of Figure 5 (Figure S3), the same peptide spectra as Figure 5 obtained from other sample positions (Figure S4), the spectra of Leu-enkephalin (YGGFL), labeled Leu-enkephalin (YGG″F″L), DRVYIHAF and labeled DRVY″I″HAF (Figures S5 and S6, respectively), and all of the major secondary ions for nonlabeled and labeled YGGFL and DRVYIHAF (Tables S1 and S2) (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: aoyagi@st.seikei.ac.jp. Fax: +81-422-37-3871.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank Prof. Ian S. Gilmore from NPL for his useful comments on the analysis and Mr. Ichiro Mihara from Kuraray for assisting in the ToF-SIMS measurement of the samples used in this study.

■ REFERENCES

(1) Ninomiya, S.; Nakata, Y.; Ichiki, K.; Seki, T.; Aoki, T.; Matsuo, J. *Nucl. Instrum. Methods Phys. Res., Sect. B* **2007**, *256* (1), 493−496.
(2) Rabbani, S.; Barber, A. M.; Fletcher, J. S.; Lockyer, N. P.; Vickerman, J. C. *Anal. Chem.* **2011**, *83* (10), 3793−3800.
(3) Mochiji, K.; Hashinokuchi, M.; Moritani, K.; Toyoda, N. *Rapid Commun. Mass Spectrom.* **2009**, *23* (5), 648−652.
(4) Yokoyama, Y.; Kawashima, T.; Ohkawa, M.; Iwai, H.; Aoyagi, S. *Surf. Interface Anal.* **2015**, *47* (4), 439−446.
(5) Shard, A. G.; Havelund, R.; Spencer, S. J.; et al. *J. Phys. Chem. B* **2015**, *119* (33), 10784−10797.
(6) Aoyagi, S.; Fletcher, J. S.; Sheraz, S.; Kawashima, T.; Berrueta Razo, I.; Henderson, A.; Lockyer, N. P.; Vickerman, J. C. *Anal. Bioanal. Chem.* **2013**, *405*, 6621−6628.
(7) Gnaser, H.; Fujii, M.; Nakagawa, S.; Seki, T.; Aoki, T.; Matsuo, J. *Rapid Commun. Mass Spectrom.* **2013**, *27*, 1490.
(8) Leufgen, K.; Mutter, M.; Vogel, H.; Szymczak, W. *J. Am. Chem. Soc.* **2003**, *125*, 8911−8915.
(9) Solé-Domenech, S.; Johansson, B.; Schalling, M.; Malm, J.; Sjovall, P. *Anal. Chem.* **2010**, *82*, 1964−1974.
(10) Aoyagi, S.; Kawashima, T.; Yokoyama, Y. *Rapid Commun. Mass Spectrom.* **2015**, *29* (18), 1687−1695.
(11) Seah, M. P.; Havelund, R.; Gilmore, I. S. *J. Phys. Chem. C* **2014**, *118*, 12862−12872.
(12) Sheraz née Rabbani, S.; Razo, I. B.; Kohn, T.; Lockyer, N. P.; Vickerman, J. C. *Anal. Chem.* **2015**, *87*, 2367−2374.
(13) Seah, M. P. *J. Phys. Chem. C* **2013**, *117*, 12622−12632.
(14) Shen, K.; Wucher, A.; Winograd, N. *J. Phys. Chem. C* **2015**, *119*, 15316−15324.
(15) Seah, M. P.; Spencer, S. J.; Havelund, R.; Gilmore, I. S.; Shard, A. G. *Analyst* **2015**, *140*, 6508−6516.

DOI: 10.1021/acs.analchem.5b04133
Anal. Chem. 2016, 88, 3592−3597